

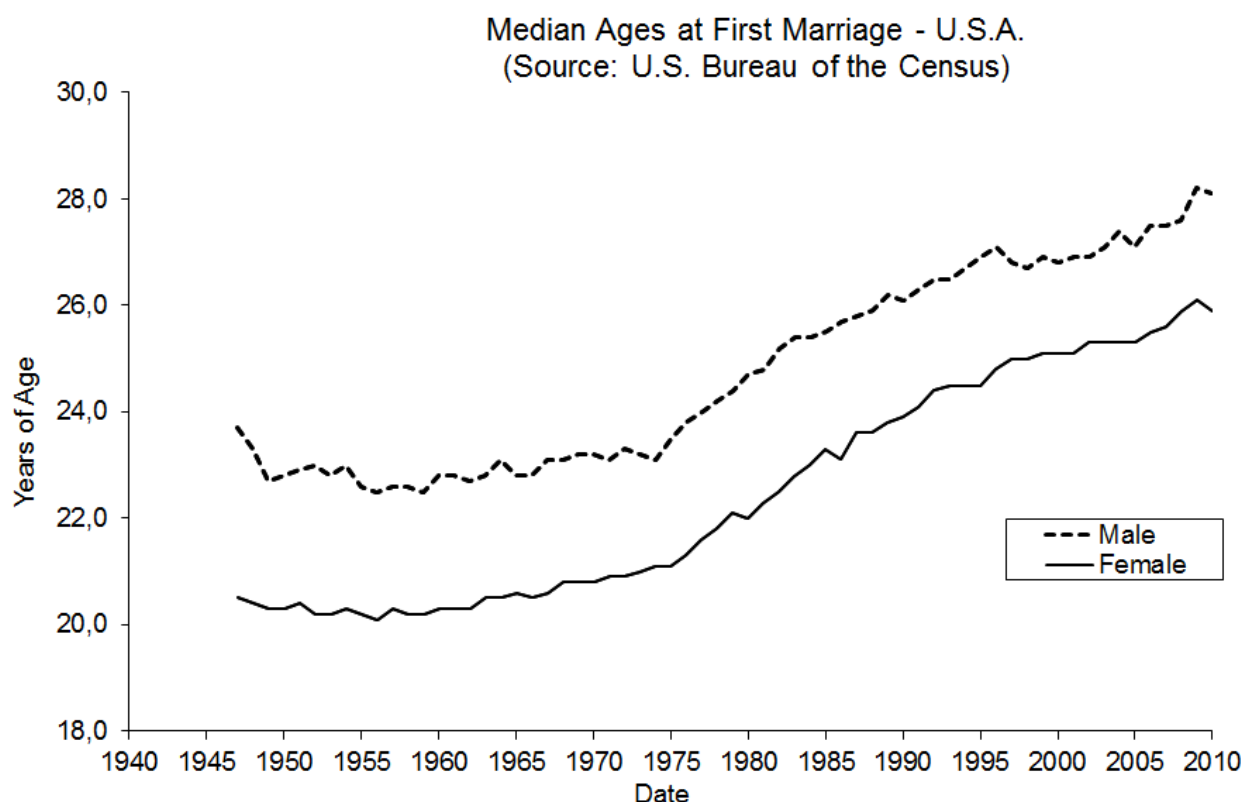
Advanced Econometrics – Exam

September 3, 2018

Exercise 1

Dave Giles at May 30, 2011 published a dataset with estimated median age of Females and Males at first marriage in the USA. In the dataset median age for each year from 1947 to 2010 is stored. The blogger suggested that possible cointegration between the median age for Females (Female) and Males (Male) may exist. To assess that, a number of analyses were performed. Furthermore, it was decided to check whether an Error Correction Mechanism model is appropriate here. Based on the results presented below please answer the following questions:

- Looking at the figure presented below, is there any rationality to consider a possibility that the Female and Male variables are cointegrated?
- Do time series of Female and Male variables satisfy basic assumptions to consider a cointegration between them (please support your answer with appropriate tests results)?
- Is there enough evidence to support a claim that there is cointegration between the variables (please support your answer with appropriate tests results)?
- What is or would be a cointegrating vector for the variables? How to interpret it?
- Does ECM model work? How would you interpret coefficients in the ECM model?



Female median age at first marriage						
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4
0	1.84	0.99	0.25	0.19	0.22	0.10
1	1.95	0.99	0.99	0.33	0.30	0.17
2	1.29	0.99	0.77	0.94	0.80	0.70
3	0.94	0.99	0.79	0.84	0.89	0.63
4	0.49	0.98	0.83	0.63	0.68	0.81
5	0.20	0.97	0.55	0.76	0.86	0.90
						0.11
						0.18
						0.44
						0.49
						0.47
						0.87

First differences for female median age at first marriage						
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4
0	-8.45	0.01	0.92	0.16	0.13	0.07
1	-4.67	0.01	0.66	0.84	0.73	0.68
2	-3.42	0.02	0.72	0.72	0.73	0.56
3	-2.29	0.21	0.79	0.56	0.57	0.69
4	-2.19	0.25	0.54	0.74	0.84	0.87
5	-1.88	0.36	0.79	0.90	0.97	0.99
						0.09
						0.40
						0.46
						0.43
						0.86
						0.99

Male median age at first marriage						
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4
0	1.33	0.99	0.86	0.88	0.28	0.43
1	1.21	0.99	0.33	0.55	0.15	0.21
2	1.32	0.99	0.94	0.96	0.06	0.11
3	0.82	0.99	0.92	0.94	0.89	0.95
4	0.67	0.99	0.86	0.98	0.98	0.98
5	0.56	0.99	0.81	0.96	0.99	0.97
						0.56
						0.32
						0.18
						0.98
						1.00
						0.99

First differences for male median age at first marriage						
augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4
0	-7.90	0.01	0.33	0.59	0.11	0.14
1	-6.82	0.01	0.89	0.98	0.04	0.07
2	-4.08	0.01	0.86	0.95	0.91	0.97
3	-3.00	0.04	0.84	0.98	0.99	0.98
4	-2.09	0.29	0.82	0.96	0.99	0.96
5	-2.19	0.25	0.98	0.79	0.91	0.96
						0.23
						0.13
						0.99
						0.99
						0.99
						0.98

Results for the Long-term relationship model

```

=====
Dependent variable:
-----
Female
-----
Male                1.102***
                   (0.017)

Constant            -4.808***
                   (0.417)

-----
Observations                64
R2                          0.986
Adjusted R2                 0.986
Residual Std. Error    0.249 (df = 62)
F Statistic             4,291.233*** (df = 1; 62)
=====
Note:                *p<0.1; **p<0.05; ***p<0.01

```

Residuals from Long-term relationship model

augmentations	adf	p_adf	p_bg.p.value.1	p_bg.p.value.2	p_bg.p.value.3	p_bg.p.value.4	p_bg.p.value.5
0	-5.38	0.01	0.57	0.72	0.42	0.59	0.73
1	-4.28	0.01	0.95	0.97	0.70	0.79	0.88
2	-2.82	0.07	0.94	0.99	0.95	0.91	0.95
3	-2.56	0.11	0.96	1.00	1.00	0.99	0.99
4	-2.11	0.28	0.94	0.93	0.98	0.99	0.99
5	-2.27	0.22	0.80	0.97	0.99	0.99	0.98

Results for the ECM model

```

=====
Dependent variable:
-----
dFemale
-----
lresid                -0.270***
                   (0.070)

dMale                0.307***
                   (0.082)

Constant            0.068***
                   (0.017)

-----
Observations                63
R2                          0.255
Adjusted R2                 0.230
Residual Std. Error    0.125 (df = 60)
F Statistic             10.280*** (df = 2; 60)
=====
Note:                *p<0.1; **p<0.05; ***p<0.01

```


Exercise 2

Data from Long (1990) on the number of publications produced by Ph.D. biochemists is analysed. In the data set the following variables are included:

- art (dependent variable – number of articles in last three years of Ph.D. produced);
- fem (gender; two levels - Female and Male);
- mar (marital status; two levels – Married and Not married);
- kid5 (number of children under age of six);
- phd (score of prestige of Ph.D. program);
- ment (number of articles produced by mentor in last three years)

Based on results presented below please answer to the following questions:

- Looking at the figure presented below, which distribution suits better to the empirical data? Are there any possible issues that are not addressed by the considered distributions?
- Is there enough evidence to support a claim about overdispersion? (please support your answer with appropriate tests results and/or measures)?
- Is there enough evidence to support a claim that there is an excess zero issue? (please support your answer with appropriate tests results)?
- Which model, among considered should be chosen as the best one (please support your answer with appropriate tests results and/or measures)?
- How would you interpret the coefficients in the Negative Binomial model?
- How would you interpret the coefficients in the Zero-Inflated Negative Binomial model?
- Your older colleague suggested that number of articles produced by a Ph.D. biochemists may have an influence on number of articles produced by a mentor, as in some of them the mentor is the co-author. What issue may be caused by such an simultaneous dependency? How would try to address this issue?

LR test for overdispersion:

Chi-square = 180.196

df = 1

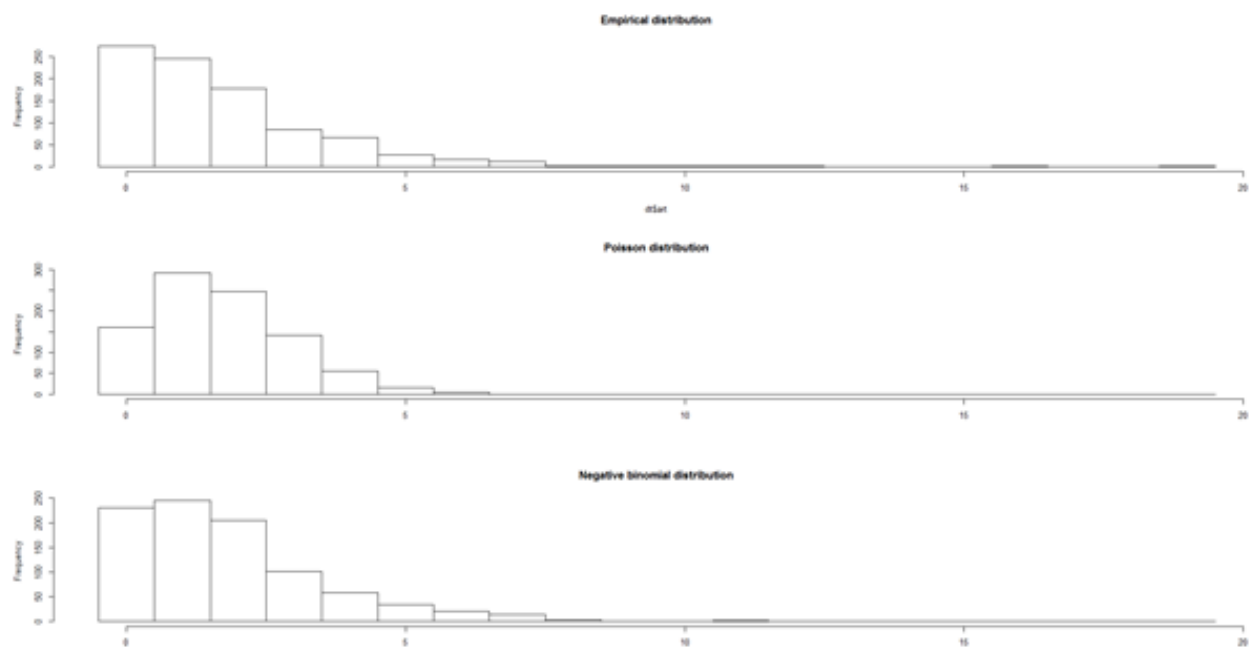
pvalue: < 0.000000000000000222

Score test for zero inflation

Chi-square = 133.91825

df = 1

pvalue: < 0.000000000000000222



Count model coefficients

=====				
Dependent variable:				
	art			
	ZIP	ZINB	Poisson	NegBin
	(1)	(2)	(3)	(4)

femMale	0.209*** (0.063)	0.196*** (0.076)	0.225*** (0.055)	0.216*** (0.074)
marNot married	-0.104 (0.071)	-0.098 (0.084)	-0.155** (0.061)	-0.150* (0.084)
kid5	-0.143*** (0.047)	-0.152*** (0.054)	-0.185*** (0.040)	-0.176*** (0.054)
phd	-0.006 (0.031)	-0.001 (0.036)	0.013 (0.026)	0.015 (0.037)
ment	0.018*** (0.002)	0.025*** (0.003)	0.026*** (0.002)	0.029*** (0.003)
Constant	0.535*** (0.112)	0.319** (0.130)	0.235** (0.093)	0.190 (0.127)
=====				

Zero-inflation model coefficients

	ZIP	ZINB		
	(1)	(2)		
femMale	-0.110 (0.280)	-0.636 (0.849)		
marNot married	0.354 (0.318)	1.499 (0.939)		
kid5	0.217 (0.196)	0.628 (0.443)		
phd	0.001 (0.145)	-0.038 (0.308)		
ment	-0.134*** (0.045)	-0.882*** (0.316)		
Constant	-0.821* (0.480)	-1.055 (1.018)		
Observations	915	915	915	915
Log Likelihood	-1,604.773	-1,549.991	-1,651.056	-1,560.958
Akaike Inf. Crit.	3245.546	3138.982	3320,113	3142.917
Note:	*p<0.1; **p<0.05; ***p<0.01			

Exercise 3

Kieran O'Connor and Amar Cheema in their paper "Do Evaluations Rise With Experience?" verified the hypothesis that professors gave higher grades the longer they offered the same course. Although the paper is publicly available, the data on courses' grades is confidential. Therefore, your lecturers estimated similar models with a different dataset.

The dataset for your exercise consisted of students' grades from a different university than the one of O'Connor and Cheema. Information on all courses given since Fall 2001 were collected. Some of the courses were offered since Fall 2001, some since Spring 2001, and so on. The courses differ in the length of their offerings. We removed from the database all the courses that were offered fewer than 4 times.

The table below presents results of five panel data models. Column (1) consists of fixed effects model, while columns (2)-(5) comprise random effects models' estimates. The dependent variable is *AvScore* that codes the average score of the given semester course. For a given semester students' results (on the 0-100 scale) were averaged. The set of independent variables consists of *Noffered* that assigns a consecutive number to each consecutive semester that the course is offered, *Sex* that codes the sex of a lecturer (0 for males, 1 for females), *Semester* – a dummy variable indicating a semester (0 Fall, 1 Spring), and *Year* included to account for the passage of time.

The R output below comes from the model with the fixed effects estimator and the Hausman test for models (1) and (2). The model had the form

$$AvScore_{it} = \beta_1 Noffered_{it} + \beta_2 Sex_{it} + u_i + \varepsilon_{it} \quad (1)$$

- a) Is the panel data in the fixed effects model balanced?
- b) Why was parameter for *Sex* not estimated in model (1)?
- c) Let's assume that we believe that the unobserved time-invariant individual effect is attributed to a positive attitude towards other people. We suppose that lecturers with a more positive attitude mark their students with higher marks. Should we in this case use the fixed or random effects estimator?
Hint: Is $E(u_i, x_{it})$ equal to zero?
- d) Interpret the Hausman test's results.
- e) What can go wrong if we decided to use simple regression for this data?
- f) Some argue that when it is green outside and the sky is blue people are in better mood. Verify the hypothesis that professors gave higher grades in Spring semesters.
- g) Some argue that due to better and better education in high schools quality of students increase over time. Does student quality increase each year? Explain.
- h) O'Connor and Cheema formulated a hypothesis that courses that award higher grades are more likely to be offered again. Results from column (5) were obtained for the random effects model estimated with courses offered at least 10 times. Comment on model (5) results.

Please don't be discouraged by the fact that dr Chlebus and dr Woźniak gave this course for the first time :-)

	<i>Dependent variable:</i>				
	AvScore				
	(1)	(2)	(3)	(4)	(5)
Noffered	0.499*** (0.002)	0.499*** (0.002)	0.499*** (0.002)	0.504*** (0.052)	0.476*** (0.091)
Sex		-0.648 (0.620)	-0.648 (0.621)	-0.651 (0.623)	-0.313 (0.983)
Semester			0.088*** (0.021)	0.088*** (0.021)	0.151*** (0.030)
Year				-0.010 (0.104)	0.048 (0.183)
Constant		83.424*** (0.445)	83.381*** (0.446)	104.026 (208.854)	-14.468 (366.431)
Observations	3,131	3,131	3,131	3,131	1,504
R ²	0.967	0.967	0.967	0.967	0.967
Adjusted R ²	0.965	0.967	0.967	0.967	0.967
F Statistic	85,625.060***	45,387.610***	30,427.620***	22,813.790***	10,845.400***

Note:

*p<0.1; **p<0.05; ***p<0.01

Oneway (individual) effect Within Model

Call:

```
plm(formula = AvScore ~ as.numeric(Noffered) + Sex, data = courses,
     model = "within", index = c("Id", "Noffered"))
```

Unbalanced Panel: n = 200, T = 4-30, N = 3131

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-1.2738347	-0.4780633	-0.0096102	0.4804250	1.3106235

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
Noffered	0.4992405	0.0017061	292.62	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

F-statistic: 85625.1 on 1 and 2930 DF, p-value: < 2.22e-16

Hausman Test

data: AvScore ~ as.numeric(Noffered) + Sex

chisq = 0.18577, df = 1, p-value = 0.0065

alternative hypothesis: one model is inconsistent

